# EXPERIMENT 6a: EXPLORATORY DATA ANALYSIS

## LEARNING OBJECTIVES:

- Understand and perform Exploratory Data Analysis (EDA) on real-world datasets.
- Load, clean, and explore the Netflix Titles dataset.
- Analyze Movies and TV Shows distribution.
- Study ratings, countries, genres, directors, and actors.
- Identify release and Netflix content addition trends.
- Analyze movie durations and TV show seasons.
- Visualize frequently occurring genres and ratings.
- Perform text analysis using descriptions and word clouds.
- Analyze monthly content addition patterns.
- Study genre co-occurrence using correlation heatmaps

### METHODS / PYTHON LIBRARIES USED:

| Library/method | Purpose | Reason for use |
|---|---|---|
| Seaborn | Statistical visualization | Creates attractive statistical plots and heatmaps |
| WordCloud | Text visualization | Displays frequently used words visually |
| Data Cleaning | Splits text into individual words/tokens | Improves data quality |
| GroupBy | Groups data by categories | Helps compare content types, countries, ratings, and years |
| Counter | Counts POS category frequencies | Helps determine the frequency of major POS categories |

## PROGRAM CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
from wordcloud import WordCloud

df = pd.read_csv("netflix_titles.csv")

print(df.shape)
print(df.info())
print(df.isnull().sum())

df = df.drop_duplicates()
df["date_added"] = pd.to_datetime(df["date_added"], errors="coerce")
df["year_added"] = df["date_added"].dt.year
df["month"] = df["date_added"].dt.month_name()

print("\nMovies vs TV Shows:")
print(df["type"].value_counts())


print("\nTop Countries:")
print(df["country"].dropna().str.split(", ").explode().value_counts().head(10))

print("\nRelease Years:")
print(df["release_year"].value_counts().head(10))

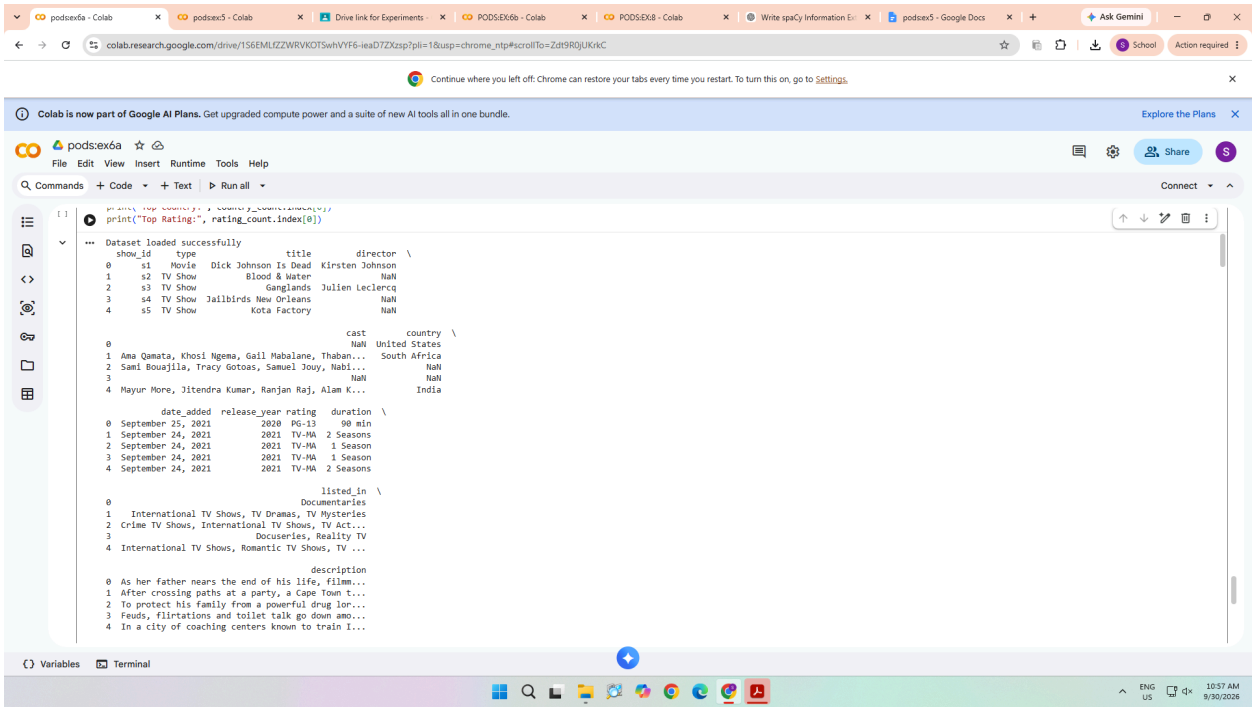
df["release_year"].value_counts().sort_index().plot()
plt.title("Release Trend")
plt.show()

print("\nContent Added by Year:")
print(df["year_added"].value_counts().sort_index())
print("\nContent Added by Month:")

print(df.sort_values("release_year", ascending=False)[["title", "release_year"]].head())

genres = df["listed_in"].dropna().str.get_dummies(", ")
plt.figure(figsize=(10, 8))
sns.heatmap(genres.corr())
plt.title("Genre Correlation Heatmap")
plt.show()
```

# OUTPUT:



```
print("Top Rating:", rating_count.index[0])

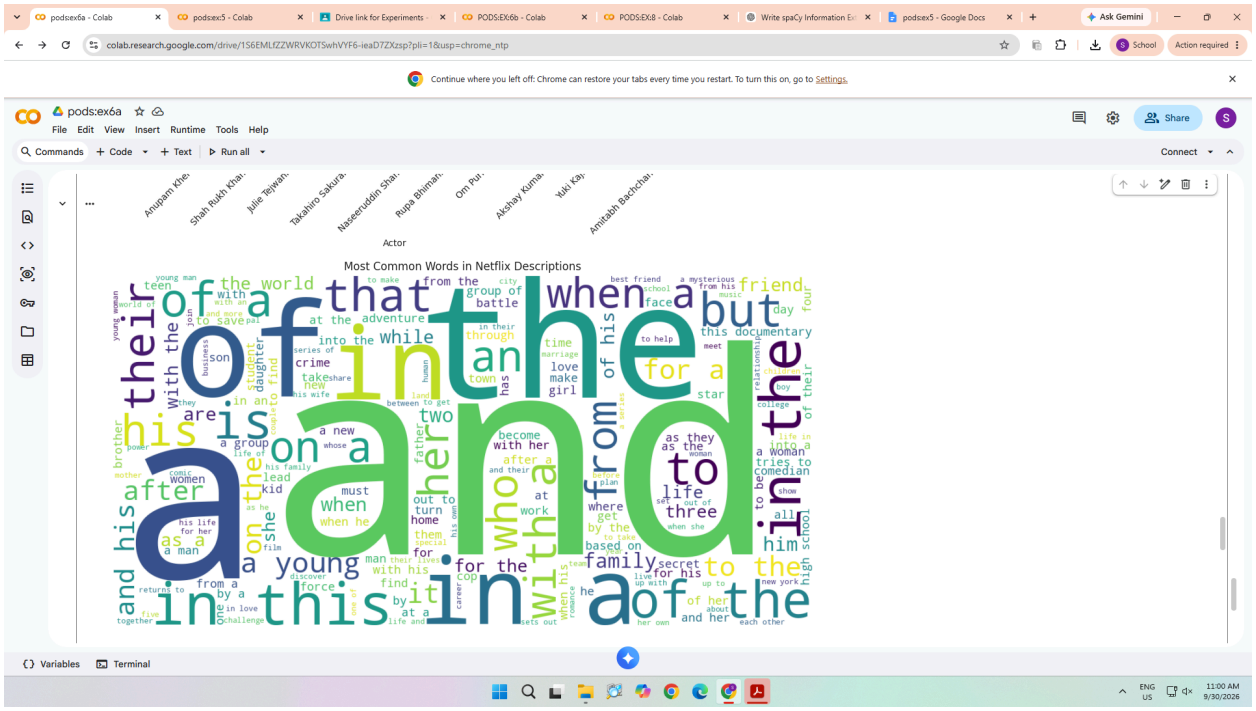
Dataset loaded successfully
  show_id     type                  title         director  \
0      s1    Movie   Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show           Blood & Water              NaN
2      s3  TV Show               Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans              NaN
4      s5  TV Show           Kota Factory              NaN

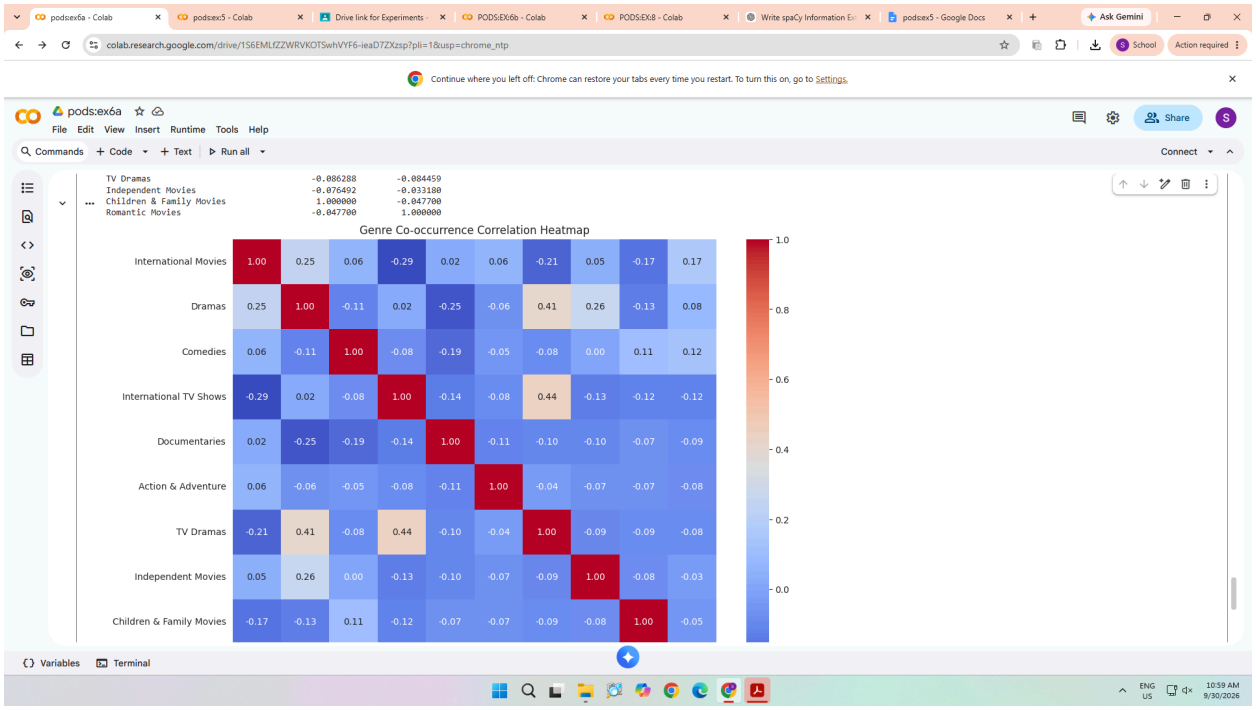
                                        cast        country  \
0                                        NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...   South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...            NaN
3                                        NaN            NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...         India

          date_added  release_year rating   duration  \
0  September 25, 2021          2020  PG-13     90 min
1  September 24, 2021          2021  TV-MA  2 Seasons
2  September 24, 2021          2021  TV-MA   1 Season
3  September 24, 2021          2021  TV-MA   1 Season
4  September 24, 2021          2021  TV-MA  2 Seasons

                                      listed_in  \
0                                  Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
3                        Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

                                         description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...
```






# LEARNING OUTCOME:

- Perform Exploratory Data Analysis on a real-world dataset.
- Clean and preprocess Netflix content data.
- Analyze Movies and TV Shows using statistical techniques.
- Identify trends in Netflix releases and content additions.
- Analyze ratings, countries, genres, directors, and actors.
- Visualize movie durations and TV show seasons.
- Perform basic text analysis using word clouds.
- Analyze monthly content addition patterns.